

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTITC404 - DESIGN AND ANALYSIS OF ALGORITHMS

Time : 3-Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 Which of the following is not one of the properties of algorithm?

1. Multiplicity
2. Ambiguity
3. Finiteness
4. Range of input

Which one of the following helps in calculating the largest amount of time taken for the completion of the algorithm?

1. Theta notation
2. Big-Oh notation
3. Omega notation
4. Alpha notation

3 The solution of $T(n) = 9T(n/3) + n^3$ recurrence relation using Master's method is

1. $\Theta(n)$
2. $\Theta(n^2)$
3. $\Theta(n^3)$
4. $\Theta(n \log n)$

The correct order of following functions by growth rate is: $n, n^2, n \log n, 2^n, n^3, n!$ is _____

1. $n, n^2, n \log n, 2^n, n^3, n!$
2. $n^3, n, n^2, n \log n, n!$
3. $n, n \log n, n^2, n^3, 2^n, n!$
4. $n^2, n \log n, 2^n, n^3, n!, n$

5 Time complexity of the Fibonacci sequence using Dynamic Programming (bottom-up) is _____

1. $O(2^n)$
2. $O(n^2)$
3. $O(n)$
4. $O(n \log n)$

6 In the 0/1 Knapsack problem, we _____

1. Take full item or leave it
2. Take fractions of item
3. Always take heavier items
4. Ignore weights

Multistage graph typically used for _____

1. Sorting data
2. Finding the longest path
3. Finding the shortest path through stages
4. Balancing a binary tree

8 In Depth First Search (DFS), nodes are typically visited using _____

1. Queue
2. Heap
3. Stack
4. Priority Queue

9 Which of the following algorithms is used to find an MST?

1. Dijkstra's Algorithm
2. Bellman-Ford Algorithm
3. Prim's Algorithm
4. Floyd-Warshall Algorithm

10 _____ problems is commonly solved using backtracking.

1. Binary Search
2. Merge Sort
3. N-Queens Problem
4. Dijkstra's Algorithm

11 What does the class P represent in computational theory?

1. Problems solved in exponential time
2. Problems that cannot be solved
3. Problems solvable in polynomial time
4. Problems verifiable in polynomial time

12 _____ is the worst case time complexity of the quick sort algorithm.

1. $O(n)$
2. $O(1)$
3. $O(n \log n)$
4. $O(n^2)$

Q2. Solve the following.

A) Define Big O notation, Big Omega and Big Theta notation. Depict the same graphically and explain.

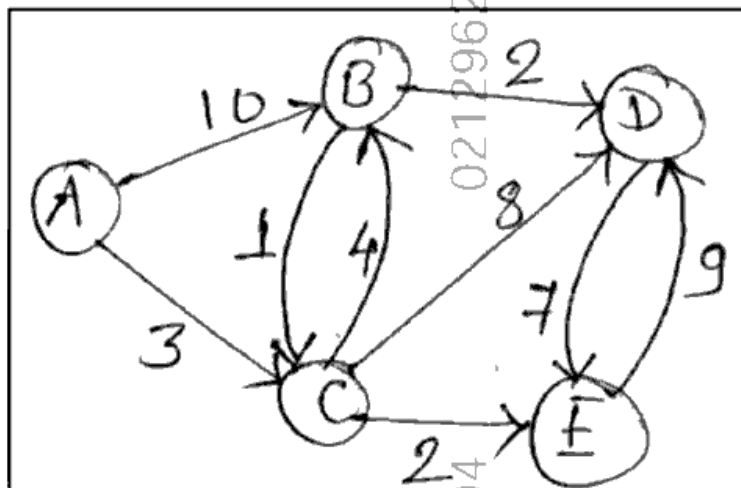
- B) Order the following functions by growth rate: n , n^2 , $n \log n$, 2^n , n^3 , $n!$. Also Solve following recurrence relation using Master's method: 6

1. $T(n) = 8T(n/2) + n^2$
2. $T(n) = 3T(n/3) + n^3$

Q3. Solve the following.

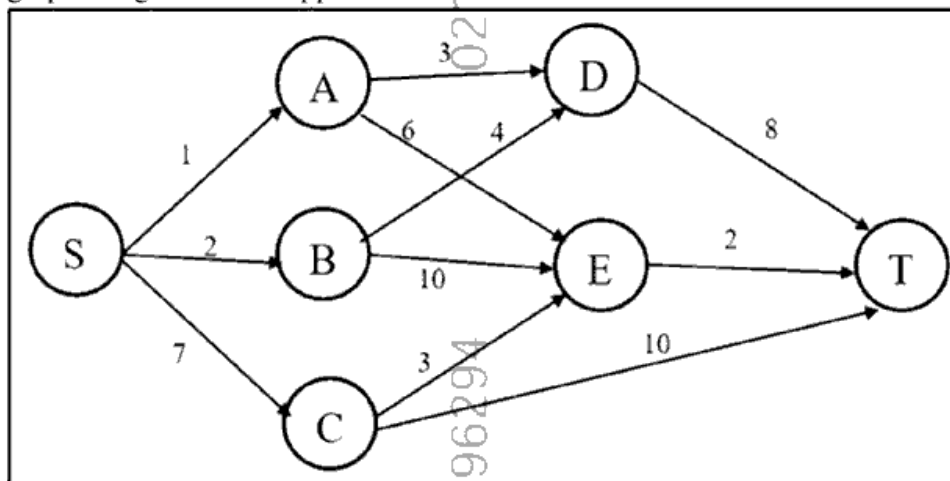
- A) Formulate a greedy approach to obtain an optimal merge pattern to pair wise merge n sorted files. Write an algorithm to generate two-way merge tree. 6

- B) Compute single source shortest path using Dijkstra's algorithm. Assuming 'A' as source vertex. 6

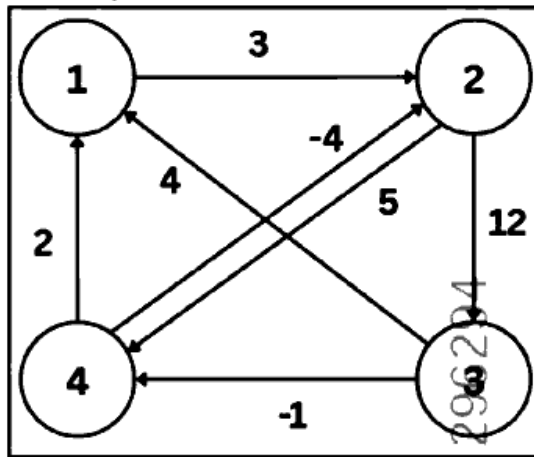


Q4. Solve Any Two of the following.

- A) Obtain the shortest path from source (S) to target (T) of the following multistage graph using backward approach. 6



- B) Trace the Floyd-Warshall's algorithm for the following graph and obtain all pair shortest path. 6



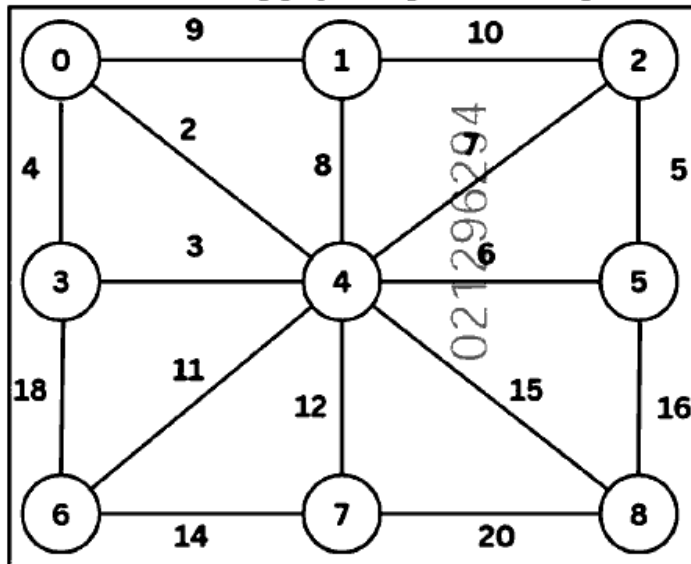
021296294

021296294

Solve following knapsack instance using dynamic programming : $n=3$, $(w_1, w_2, w_3) = (2, 3, 4)$ and $(p_1, p_2, p_3) = (1, 2, 5)$. Where $M=6$. 6

Q5. Solve Any Two of the following.

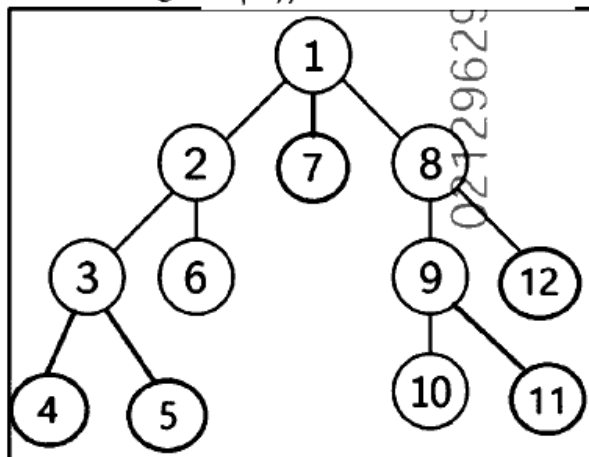
- A) Define Spanning tree, minimum spanning tree. Obtain the minimum cost spanning tree for the following graph using Kruskal's algorithm. 6



021296294

021296294

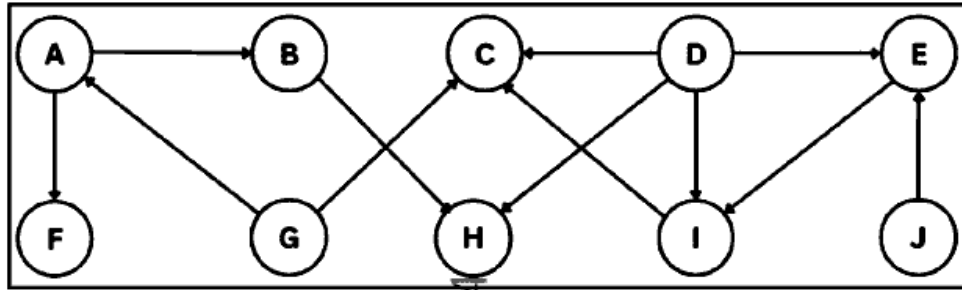
- B) Differentiate between BFS and DFS traversal. Write DFS and BFS traversal for the following. <https://www.batuonline.com> 6



021296294

021296294

- C) Explain topological sort in detail. Apply topological sort and obtain ordering for following graph. 6



Q6. Solve Any Two of the following.

- A) Solve the following sum of subset problem using backtracking and draw a state space tree. $m = 8$ and $w = \{1, 3, 4, 5\}$ 6
- B) Differentiate between backtracking and branch & bound technique. 6
- C) Draw and explain P, NP, NP-Complete and NP-hard Problem. 6

*** End ***